

MIDDLETOWN, OH, USA

WMO: 722249

Lat: 39.531N

Lon: 84.395W

Elev: 650

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 63887

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	10.4	-5.2	4.3	9.3	0.3	5.7	14.0	24.2	38.3	20.9	34.0	3.1	230	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	94.6	76.6	90.6	74.3	89.8	73.8	79.4	89.5	77.5	87.6	75.7	84.1	8.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	143.4	84.5	74.8	133.6	81.4	73.1	125.7	80.1	43.4	89.3	41.3	87.9	39.6	85.1	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	18.0	15.9	-2.6	95.9	8.0	2.6	-8.3	97.8	-13.0	99.4	-17.5	100.8	-23.3	102.8		
			-2.9	82.3	7.5	2.0	-8.3	83.8	-12.7	84.9	-16.9	86.1	-22.4	87.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	30.6	32.7	43.5	54.6	65.2	72.6	75.3	74.8	68.9	56.6	44.0	36.7
	DBStd	18.29	12.07	11.27	10.46	9.12	7.74	5.74	5.29	5.38	7.42	9.27	8.86	9.83
	HDD50	2102	608	493	258	56	2	0	0	0	0	38	219	428
	HDD65	4915	1066	904	669	325	96	9	1	1	44	292	629	879
	CDD50	3838	8	9	56	195	473	678	785	769	568	242	41	14
	CDD65	1172	0	0	1	14	102	236	320	306	161	32	0	0
	CDH74	11228	0	2	23	210	1011	2181	3026	2871	1563	337	4	0
	CDH80	4250	0	0	1	28	283	826	1227	1150	640	95	0	0
Wind	WSAvg	5.7	7.4	7.2	6.9	7.3	5.5	5.1	4.2	3.9	4.0	5.3	5.6	6.4
Precipitation	PrecAvg	40.7	3.2	2.1	3.3	3.9	4.8	4.4	4.0	3.1	2.9	3.0	2.8	3.2
	PrecMax	57.0	7.5	4.6	7.7	8.9	9.1	7.6	8.5	5.0	6.2	6.3	5.0	8.7
	PrecMin	30.2	1.3	0.8	1.1	1.4	1.3	1.1	2.4	1.5	0.7	0.9	0.8	1.0
	PrecStd	6.3	1.4	1.1	1.4	1.8	2.2	1.6	1.6	1.1	1.7	1.6	1.1	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.8	68.6	77.2	82.3	90.4	95.0	96.7	97.2	95.4	90.6	74.9	68.1
		MCWB	60.7	58.3	62.0	63.4	73.1	77.5	79.1	80.1	72.5	72.6	61.0	61.8
	2%	DB	59.6	60.9	72.1	80.1	86.8	90.8	92.5	93.2	90.9	81.3	69.2	60.7
		MCWB	55.0	54.1	61.2	62.6	71.2	73.8	77.6	75.1	72.4	65.6	58.4	55.8
	5%	DB	54.7	55.2	66.2	75.9	82.4	89.6	90.2	90.2	88.9	77.2	63.8	56.8
		MCWB	51.3	49.5	55.0	60.7	68.6	72.7	75.4	74.7	72.2	64.3	56.7	53.3
	10%	DB	48.9	49.6	60.9	72.3	80.2	84.7	88.0	87.8	82.3	73.0	59.6	53.3
		MCWB	44.6	44.4	52.2	59.9	67.4	70.9	75.3	72.8	67.9	63.2	51.4	49.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.4	61.8	66.0	69.2	76.0	79.8	82.4	82.4	77.3	73.6	65.9	62.3
		MCDB	65.6	68.1	74.9	77.6	86.9	90.4	91.7	92.8	86.7	85.6	70.7	66.0
	2%	WB	55.4	56.1	61.4	66.1	73.2	77.2	79.8	78.9	75.3	70.7	61.3	57.2
		MCDB	57.9	59.5	69.9	74.1	82.2	88.5	89.4	89.5	84.7	77.8	66.5	59.8
	5%	WB	51.4	50.8	57.6	63.7	71.6	75.3	78.0	76.8	73.5	67.5	56.7	53.7
		MCDB	54.7	55.2	63.1	72.0	79.8	84.1	87.5	86.1	82.7	73.3	62.5	55.9
	10%	WB	45.0	44.6	54.1	61.2	69.8	73.5	76.0	74.8	71.9	64.7	53.5	49.5
		MCDB	48.1	49.7	59.4	69.3	77.6	81.2	83.5	82.9	80.6	71.4	58.6	53.1
Mean Daily Temperature Range	5% DB	MDBR	15.6	16.7	19.5	22.9	21.5	21.0	20.1	21.3	22.6	21.8	20.1	15.0
		MCDBR	20.7	24.8	27.9	28.3	25.8	25.8	23.4	24.4	28.2	26.6	27.1	20.7
		MCWBR	17.5	19.3	17.0	15.3	12.7	11.9	10.7	10.6	11.7	13.3	18.1	18.0
	5% WB	MCDBR	19.3	22.9	22.3	23.1	23.2	22.2	20.8	21.4	22.9	21.3	22.9	19.0
MCWBR		17.3	19.4	16.1	15.3	12.3	11.8	10.7	10.5	10.3	13.1	17.6	17.8	
Clear-Sky Solar Irradiance	taub	0.305	0.328	0.352	0.398	0.442	0.453	0.456	0.451	0.400	0.358	0.328	0.305	
	taud	2.455	2.359	2.366	2.252	2.170	2.189	2.187	2.183	2.302	2.419	2.462	2.469	
	Ebn at Noon	275	282	286	278	267	263	261	259	266	268	263	266	
	Edh at Noon	25	32	35	42	47	46	46	45	37	30	25	23	
All-Sky Solar Radiation	RadAvg	554	814	1140	1539	1741	1930	1876	1734	1423	1013	685	468	
	RadStd	47	97	98	124	145	121	113	88	137	96	79	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+133

Nomenclature: See separate page